MSG No. 135.493 P_C4_2/mbc [Type IV, tetragonal]

Table 1: Wyckoff site: `4a`, site symmetry: `mm'm'.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,6,9,14,20,21,28,29]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[2,3,10,11,23,24,31,32]`
3	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[4,5,12,13,17,22,25,30]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[7,8,15,16,18,19,26,27]`

Table 2: Wyckoff site: `4b`, site symmetry: `-4'm'2`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,6,7,8,26,27,28,29]`
2	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[2,3,4,5,25,30,31,32]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[9,14,15,16,18,19,20,21]`
4	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[10,11,12,13,17,22,23,24]`

Table 3: Wyckoff site: `4c`, site symmetry: `mm'm'.`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,6,9,14,20,21,28,29]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[2,3,10,11,23,24,31,32]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[4,5,12,13,17,22,25,30]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[7,8,15,16,18,19,26,27]`

Table 4: Wyckoff site: `4d`, site symmetry: `-4m'2'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,6,10,11,23,24,28,29]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[2,3,9,14,20,21,31,32]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{3}{4}\right]$	`[4,5,15,16,18,19,25,30]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{4}\right]$	`[7,8,12,13,17,22,26,27]`

Table 5: Wyckoff site: `8e`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[1,8,22,23]`
2	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{3}{4}\right]$	`[2,4,19,21]`
3	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[3,5,18,20]`
4	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[6,7,17,24]`
5	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{3}{4}\right]$	`[9,16,30,31]`
6	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{4}\right]$	`[10,12,27,29]`
7	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{4}\right]$	`[11,13,26,28]`

8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[14,15,25,32]

Table 6: Wyckoff site: 8f, site symmetry: 2'/m..

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	[1,14,22,25]
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[2,11,19,26]
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[3,10,18,27]
4	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	[4,13,21,28]
5	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	[5,12,20,29]
6	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	[6,9,17,30]
7	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	[7,16,24,31]
8	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	[8,15,23,32]

Table 7: Wyckoff site: 8g, site symmetry: 2m'm'.

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	[1,6,28,29]
2	$\left[\frac{1}{2}, 0, z+\frac{1}{2}\right]$	[2,3,31,32]
3	$\left[\frac{1}{2}, 0, -z\right]$	[4,5,25,30]
4	$\left[0, \frac{1}{2}, \frac{1}{2}-z\right]$	[7,8,26,27]
5	$\left[0, \frac{1}{2}, -z\right]$	[9,14,20,21]
6	$\left[\frac{1}{2}, 0, \frac{1}{2}-z\right]$	[10,11,23,24]
7	$\left[\frac{1}{2}, 0, z\right]$	[12,13,17,22]
8	$\left[0, \frac{1}{2}, z+\frac{1}{2}\right]$	[15,16,18,19]

Table 8: Wyckoff site: 8h, site symmetry: 2m'm'.

No.	position	mapping
1	$[0, 0, z]$	[1,6,28,29]
2	$\left[0, 0, z+\frac{1}{2}\right]$	[2,3,31,32]
3	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	[4,5,25,30]
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-z\right]$	[7,8,26,27]
5	$[0, 0, -z]$	[9,14,20,21]
6	$\left[0, 0, \frac{1}{2}-z\right]$	[10,11,23,24]
7	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	[12,13,17,22]
8	$\left[\frac{1}{2}, \frac{1}{2}, z+\frac{1}{2}\right]$	[15,16,18,19]

Table 9: Wyckoff site: 8i, site symmetry: mm'2'.

No.	position	mapping
1	$[0, y, 0]$	[1,14,21,28]
2	$\left[-y, 0, \frac{1}{2}\right]$	[2,11,24,31]
3	$\left[y, 0, \frac{1}{2}\right]$	[3,10,23,32]

continued ...

Table 9

No.	position	mapping
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[4,13,22,25]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ 0\right]$	[5,12,17,30]
6	$[0,\ -y,\ 0]$	[6,9,20,29]
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	[7,16,19,26]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}\right]$	[8,15,18,27]

Table 10: Wyckoff site: 8j, site symmetry: mm'2'.

No.	position	mapping
1	$\left[0,\ y,\ \frac{1}{2}\right]$	[1,14,21,28]
2	$[-y,\ 0,\ 0]$	[2,11,24,31]
3	$[y,\ 0,\ 0]$	[3,10,23,32]
4	$\left[\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[4,13,22,25]
5	$\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[5,12,17,30]
6	$\left[0,\ -y,\ \frac{1}{2}\right]$	[6,9,20,29]
7	$\left[y+\frac{1}{2},\ \frac{1}{2},\ 0\right]$	[7,16,19,26]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ 0\right]$	[8,15,18,27]

Table 11: Wyckoff site: 16k, site symmetry: 2'..

No.	position	mapping
1	$\left[\frac{3}{4},\ \frac{3}{4},\ z\right]$	[1,22]
2	$\left[\frac{1}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[2,19]
3	$\left[\frac{3}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[3,18]
4	$\left[\frac{1}{4},\ \frac{3}{4},\ -z\right]$	[4,21]
5	$\left[\frac{3}{4},\ \frac{1}{4},\ -z\right]$	[5,20]
6	$\left[\frac{1}{4},\ \frac{1}{4},\ z\right]$	[6,17]
7	$\left[\frac{1}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[7,24]
8	$\left[\frac{3}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[8,23]
9	$\left[\frac{1}{4},\ \frac{1}{4},\ -z\right]$	[9,30]
10	$\left[\frac{3}{4},\ \frac{1}{4},\ \frac{1}{2}-z\right]$	[10,27]
11	$\left[\frac{1}{4},\ \frac{3}{4},\ \frac{1}{2}-z\right]$	[11,26]
12	$\left[\frac{3}{4},\ \frac{1}{4},\ z\right]$	[12,29]
13	$\left[\frac{1}{4},\ \frac{3}{4},\ z\right]$	[13,28]
14	$\left[\frac{3}{4},\ \frac{3}{4},\ -z\right]$	[14,25]
15	$\left[\frac{3}{4},\ \frac{3}{4},\ z+\frac{1}{2}\right]$	[15,32]
16	$\left[\frac{1}{4},\ \frac{1}{4},\ z+\frac{1}{2}\right]$	[16,31]

Table 12: Wyckoff site: 16l, site symmetry: ..2

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[1,7]
2	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	[2,5]

continued ...

Table 12

No.	position	mapping
3	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[3,4]
4	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[6,8]
5	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[9,15]
6	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[10,13]
7	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	[11,12]
8	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[14,16]
9	$\left[x+\frac{1}{2},\ x,\ \frac{1}{4}\right]$	[17,23]
10	$\left[-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[18,21]
11	$\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[19,20]
12	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{4}\right]$	[22,24]
13	$\left[\frac{1}{2}-x,\ -x,\ \frac{3}{4}\right]$	[25,31]
14	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[26,29]
15	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[27,28]
16	$\left[x+\frac{1}{2},\ x,\ \frac{3}{4}\right]$	[30,32]

Table 13: Wyckoff site: 16m, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{4}\right]$	[1,23]
2	$\left[-x,\ x,\ \frac{3}{4}\right]$	[2,21]
3	$\left[x,\ -x,\ \frac{3}{4}\right]$	[3,20]
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[4,19]
5	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[5,18]
6	$\left[-x,\ -x,\ \frac{1}{4}\right]$	[6,24]
7	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[7,17]
8	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[8,22]
9	$\left[-x,\ -x,\ \frac{3}{4}\right]$	[9,31]
10	$\left[x,\ -x,\ \frac{1}{4}\right]$	[10,29]
11	$\left[-x,\ x,\ \frac{1}{4}\right]$	[11,28]
12	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[12,27]
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[13,26]
14	$\left[x,\ x,\ \frac{3}{4}\right]$	[14,32]
15	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[15,25]
16	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[16,30]

Table 14: Wyckoff site: 16n, site symmetry: m..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,14]
2	$\left[-y,\ x,\ \frac{1}{2}\right]$	[2,11]
3	$\left[y,\ -x,\ \frac{1}{2}\right]$	[3,10]
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ 0\right]$	[4,13]
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ 0\right]$	[5,12]
6	$[-x,\ -y,\ 0]$	[6,9]

continued ...

Table 14

No.	position	mapping
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[7,16]
8	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[8,15]
9	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, 0\right]$	[17,30]
10	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[18,27]
11	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[19,26]
12	$[x,\, -y,\, 0]$	[20,29]
13	$[-x,\, y,\, 0]$	[21,28]
14	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, 0\right]$	[22,25]
15	$\left[y,\, x,\, \frac{1}{2}\right]$	[23,32]
16	$\left[-y,\, -x,\, \frac{1}{2}\right]$	[24,31]

Table 15: Wyckoff site: 16o, site symmetry: .m'.

No.	position	mapping
1	$[0,\, y,\, z]$	[1,28]
2	$\left[-y,\, 0,\, z+\frac{1}{2}\right]$	[2,31]
3	$\left[y,\, 0,\, z+\frac{1}{2}\right]$	[3,32]
4	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[4,25]
5	$\left[\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[5,30]
6	$[0,\, -y,\, z]$	[6,29]
7	$\left[y+\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}-z\right]$	[7,26]
8	$\left[\frac{1}{2}-y,\, \frac{1}{2},\, \frac{1}{2}-z\right]$	[8,27]
9	$[0,\, -y,\, -z]$	[9,20]
10	$\left[y,\, 0,\, \frac{1}{2}-z\right]$	[10,23]
11	$\left[-y,\, 0,\, \frac{1}{2}-z\right]$	[11,24]
12	$\left[\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[12,17]
13	$\left[\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[13,22]
14	$[0,\, y,\, -z]$	[14,21]
15	$\left[\frac{1}{2}-y,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[15,18]
16	$\left[y+\frac{1}{2},\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[16,19]

Table 16: Wyckoff site: 32p, site symmetry: 1

No.	position	mapping
1	$[x,\, y,\, z]$	[1]
2	$\left[-y,\, x,\, z+\frac{1}{2}\right]$	[2]
3	$\left[y,\, -x,\, z+\frac{1}{2}\right]$	[3]
4	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, -z\right]$	[4]
5	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, -z\right]$	[5]
6	$[-x,\, -y,\, z]$	[6]
7	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[7]
8	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[8]
9	$[-x,\, -y,\, -z]$	[9]
10	$\left[y,\, -x,\, \frac{1}{2}-z\right]$	[10]

continued ...

Table 16

No.	position	mapping
11	$\left[-y,\, x,\, \frac{1}{2}-z\right]$	[11]
12	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z\right]$	[12]
13	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z\right]$	[13]
14	$[x,\, y,\, -z]$	[14]
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[16]
17	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[17]
18	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[18]
19	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[19]
20	$[x,\, -y,\, -z]$	[20]
21	$[-x,\, y,\, -z]$	[21]
22	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, z\right]$	[22]
23	$\left[y,\, x,\, \frac{1}{2}-z\right]$	[23]
24	$\left[-y,\, -x,\, \frac{1}{2}-z\right]$	[24]
25	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[25]
26	$\left[y+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-z\right]$	[26]
27	$\left[\frac{1}{2}-y,\, x+\frac{1}{2},\, \frac{1}{2}-z\right]$	[27]
28	$[-x,\, y,\, z]$	[28]
29	$[x,\, -y,\, z]$	[29]
30	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, -z\right]$	[30]
31	$\left[-y,\, -x,\, z+\frac{1}{2}\right]$	[31]
32	$\left[y,\, x,\, z+\frac{1}{2}\right]$	[32]